LETA: A Lattice-Based Efficient and Traceable Privacy-Preserving Batch Authentication Scheme for Vehicle Platoon in VANETs

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***Abstract*—Vehicle platoon (VP), as a typical form of traffic cooperation, can significantly enhance traffic efficiency and safety in Vehicular Ad hoc Networks (VANETs). However, malicious vehicles in VP poses a severe threat to the security of entire VP, requiring to be efficiently traced by identity authentication. In this paper, we propose a lattice-based efficient and traceable privacy-preserving batch authentication scheme for vehicle platoon in VANETs, named LETA. First, we design a dynamic VP identity structure VPD-Tree which is constructed based on hash tree and pseudonyms of vehicles to preserve privacy. Then, an aggregate signature is constructed based on VPD-tree and modular lattice for secure and efficient batch authentication of VP. Finally, Zero-Knowledge Proofs (ZKP) is applied on the VPD-Tree structure to anonymously and efficiently trace the malicious vehicles of VP. Security analysis shows that LETA achieves stronger security guarantees, thereby offering a more secure solution than existing approaches. Moreover, performance evaluations show that LETA achieves lower computation and communication overheads through the VPD-tree structure and efficient batch authentication scheme.**

***Index Terms*—Vehicle platoon, batch authentication, aggregation signature, privacy-preserving, efficient traceability.**

I. Introduction

With the rapid development of Vehicular Ad hoc Networks (VANETs), vehicle platoon (VP) has become an important approach for traffic collaboration which can significantly enhance traffic efficiency and safety in various application scenarios. VP refers to a group of vehicles that travel together in a coordinated manner, maintaining close distances and synchronizing their movements through inter-vehicle communication. For instance, during peak hours, a platoon of buses can effectively alleviate traffic congestion and reduce energy consumption [1]. In freight transportation, well-coordinated truck platoons can improve transportation efficiency and reduce air resistance, thereby decreasing fuel consumption [2]. In military scenarios, automated vehicle platoons can reduce unreliability of human operation [3], thereby improving the accuracy of execution. Due to the coordination and cooperation required among multiple vehicles in VP, the presence of malicious vehicles could pose a serious threat to the security of entire platoon [4]. Therefore, it is essential to ensure that all vehicles in the VP are legitimate and trusted participants, and the malicious vehicles can be efficiently and accurately traced if present.

In order to effectively verify the legitimacy of all vehicle identities in VP, the authentication scheme is required to be designed with three aspects of characteristics. (1) For verifying multiple vehicle identities of VP, it is crucial to design an efficient authentication scheme that reduces computational cost while maintaining strong security [5]. (2) During authentication for vehicles in VP, the privacy of vehicles is required to be strictly preserved to prevent identity fraud and information leakage. (3) If there existing malicious vehicles in VP, they are required to be efficiently and accurately identified and traced to maintain stability and security of the entire platoon [6].

Aiming to fulfill the characteristics above, some batch authentication schemes are proposed for efficiently verifying the legitimacy of vehicle identities within the platoon. These work primarily include methods based on bilinear pairings [7], [8] and Elliptic Curve Cryptography (ECC) [9], [10], [11]. However, with the rise of quantum attack, the security capability of these traditional methods becomes increasingly vulnerable. Therefore, researchers choose some stronger cryptographic techniques to design the authentication schemes to resist quantum attacks, such as lattice-based cryptography [12], [13], [14]. Although these lattice-based authentication schemes offer enhanced security, they come with the drawback of higher computation and communication costs.

Received 12 February 2025; revised 15 August 2025; accepted 21 November 2025. Date of publication 25 November 2025; date of current version 18 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62472132 and in part by the Key Research and Development Program Project of Zhejiang Province under Grant 2024C01179. *(Corresponding author: Yingjie Xia.)*

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Digital Object Identifier 10.1109/JSAC.2025.3636982

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Additionally, when batch authentication fails due to the presence of malicious vehicles, tracing those vehicles in the VP typically requires re-authentication of all vehicle identities [15] or verification of other auxiliary information [16]. These VP batch authentication and malicious vehicle tracing tasks not only incur higher computational and communication costs but also raise potential risks of privacy leakage, particularly in the VP with many vehicles constantly sharing sensitive information.

To address the aforementioned issues, we propose a lattice-based efficient and traceable privacy-preserving batch authentication scheme for vehicle platoon in VANETs, named LETA. Specifically, the scheme consists of three components. First, a dynamic VP integrated identity structure VPD-Tree is constructed based on the hash tree and pseudonyms of vehicles, embedding cryptographic randomness to protect against privacy leakage and prevent identity linkage attack. Second, an aggregated signature is designed through utilizing VPD-Tree and modular lattice to enable secure and efficient batch authentication for all vehicles in VP. Third, Zero-Knowledge Proofs (ZKP) is applied on the VPD-Tree structure to implement an anonymous and efficient malicious vehicles tracing method, allowing for quick identification of malicious vehicles without compromising privacy. Therefore, the main contributions can be summarized as follows.

1) The hash tree-based VPD-Tree structure is designed by the nodes constructed by pseudonyms and other information of each vehicle in VP with the partial update mechanism, enabling dynamic VP integrated identity construction, efficient VPD-Tree update and effective privacy preservation.
2) The lattice-based aggregate signature is designed by generating vehicle hash values from VPD-Tree and reducing VP signature size through modular lattice, thereby achieving secure and efficient batch authentication for all vehicles in VP.
3) The malicious vehicle tracing method leverages the hierarchical structure of VPD-Tree to quickly locate suspected malicious vehicles and uses ZKP in the tree nodes to verify vehicle identity without disclosing confidential information, thereby enabling anonymous and efficient trace in VP.

The structure of remaining parts of this paper is organized as follows. Section II reviews related work. Section III provides an overview of the LETA architecture, and Section IV details implementation of the LETA scheme. The security analysis and performance experiments are conducted in Section V and Section VI respectively. Finally, Section VII concludes the paper.

II. Relate Work

A. Authentication Schemes for Vehicle Platoons in VANETs

Authentication in VANETs, particularly for VP, is complex due to the dynamic nature of the network, the high mobility of vehicles, and the need for fast, efficient communication protocols. Several methods have been proposed to authenticate vehicles in such environments, each with its own strengths and limitations. These include traditional public key infrastructure (PKI)-based methods, identity-based cryptography (IBC), and certificateless cryptography (CLC).

Asghar et al. [17] proposed a similar PKI-based authentication protocol for VANETs. However, as the number of revoked vehicles grows, the efficiency of identifying revoked vehicles from the Certificate Revocation Lists (CRL) becomes relatively low. In general, PKI schemes suffer from high computational overhead, require complex certificate management, and are vulnerable to scalability issues in dynamic environments like VANETs. Cui et al. [18] proposed a privacy-preserving authentication scheme based on a cuckoo filter, which leveraged the cuckoo filter and binary search to enhance batch verification. Zhang et al. [19] employed bilinear pairing-based signature technology to safeguard their scheme against collusion and replay attacks. However, IBC-based schemes faced challenges such as increased computational complexity, scalability issues in large-scale networks, and potential vulnerabilities to advanced attacks like adaptive collusion. To address some of the issues with traditional PKI and IBC schemes, the use of CLC has been proposed. Yadav and Vijayakumar [20] proposed a signature-based authentication protocol for VANETs that provided an authentication solution for vehicles, Roadside Units (RSUs), and Trusted Authorities (TAs). Nath and Choudhury [11] introduced a privacy-preserving authentication scheme for VANETs, in which authenticated vehicles communicated using a group key. To prevent traceability attacks, vehicles were required to request pseudonym updates from the TA.

However, unlike traditional VANETs, VP is characterized by a structured and tightly coordinated topology. Therefore, existing authentication schemes lack the scalability required to handle the high mobility and dynamic changes inherent to VP scenarios.

B. Batch Authentication for Vehicle Platoons in VANETs

In VANETs, batch authentication improves efficiency by simultaneously verifying multiple vehicles, thereby reducing both computational and communication overhead. Existing techniques include group signatures [21], ring signatures [22], and aggregate signatures. Among them, aggregate signatures offer significant advantages by combining multiple individual signatures into a single compact one, making the process more efficient than group or ring signatures. This further optimizes the authentication workflow by enhancing speed and scalability, while also reducing communication and computational cost.

In 2003, Boneh et al. [23] proposed the concept of aggregate signatures, which combine n individual signatures into a single short signature, significantly reducing message size and authentication overhead. Zhang et al. [24] proposed an identity authentication scheme based on aggregate signatures, bilinear pairings, and temporary pseudonyms. It ensured secure vehicle authentication, message validation, and tracking prevention, while maintaining strong performance and resilience to security threats. Ali et al. [25] proposed an efficient and secure certificateless short signature-based conditional privacy-preserving authentication scheme, referred to as CLSS-CPPA,

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for V2V communication. This scheme avoided the use of bilinear pairings and map-to-point hash functions, instead relying on ECC and general hash functions. It supported batch signature verification, improving computational and communication efficiency, and ensured security against various attacks based on the ECDLP. Zhu et al. [26] proposed a certificateless conditional privacy-preserving authentication scheme for VANETs, which was based on ECC to generate aggregate signatures, enhancing the security of the authentication process while preserving user privacy. However, with the advancement of quantum computing, the security of traditional encryption schemes has become increasingly vulnerable, as quantum computers can potentially break widely used cryptographic schemes, such as RSA and ECC, through techniques like Shor's algorithm.

This has prompted researchers to explore lattice-based cryptography, which is considered as one of the most promising post-quantum cryptographic approaches. Wani et al. [27] proposed a lattice-based dynamic anonymous batch-verification authentication scheme that utilized fuzzy extractors to achieve dynamic anonymity and avoid pseudonym storage. This scheme addressed the limitations of existing VANETs authentication schemes in terms of quantum resistance and storage efficiency. Nath and Choudhury [28] introduced a lattice-based privacy-preserving mutual authentication scheme, named LbPV, which was designed to address security and privacy challenges in vehicle message exchanges in VANETs. The scheme eliminated service provider monitoring, enabled vehicles to authenticate without generating real-time parameters, and leveraged lattice-based encryption to enhance system security. Li et al. [29] proposed a lattice-based aggregate signature scheme with dynamic identity revocation, which improved authentication efficiency and security while resisting impersonation attacks in quantum computing environments. Prajapat et al. [30] introduced a lattice-based identity-based aggregate signature scheme that combined batch verification and aggregate signature schemes to efficiently authenticate multiple vehicles and aggregate their signatures. However, the signatures generated by lattice-based cryptography tend to be larger in size, which can result in higher communication and computational costs, especially in resource-constrained environments like VANETs. Moreover, most existing lattice-based authentication schemes are not well-suited for vehicular platoons, as they lack structured identity management and support for continuous batch authentication. The tightly coupled and hierarchical nature of platoons demands low-latency responses and dynamic identity updates.

C. Malicious Vehicles Tracing for Vehicle Platoon in VANETs

In batch authentication, when the aggregate signature verification fails, only a few scheme consider the malicious vehicles tracing. Most approaches rely on re-verifying all signatures to determine the source of the malicious vehicles [31], [32]. Although this scheme can effectively identify the malicious vehicles, it incurs significant cost due to the large number of computational and verification operations required, especially in scenarios involving a large number of vehicles. This inefficiency becomes particularly problematic in VP, where rapid authentication and traceability are critical for maintaining VP reliability and security. Furthermore, the sequential nature of re-verifying individual signatures limits the scalability of such methods, as the computational cost increases linearly with the number of signatures. This not only affects real-time performance but also poses challenges in scenarios where multiple malicious entities may be involved. Shen et al. [33] proposed a binary tree traversal method to efficiently track vehicles with invalid signatures, thereby reducing computational overhead. However, directly embedding signatures within the binary tree could potentially compromise the system's security and privacy. Nevertheless, this approach offers an innovative perspective for advancing batch authentication research in VP.

In VP scenarios, anonymous and efficient tracing is essential to ensure the overall security of the platoon. Traditional batch authentication schemes are designed for general VANETs and typically overlook the structured nature and frequent membership changes in VP. While they can authenticate vehicle identities, they are inefficient at tracing malicious vehicles, resulting in high computational overhead and slow response. Therefore, our work focuses on designing a tracing mechanism specifically optimized for the dynamic nature of VP, supporting efficient authentication while enabling rapid and scalable tracing of malicious vehicles.

III. System Overview

A. System Architecture

The system architecture for LETA is shown in Fig. 1. The infrastructure of the system consists of Trusted Authority(TA), Trusted edge center(TEC), Roadside Unit(RSU), Vehicle Platoon(VP), and Blockchain(BC). The specific descriptions of these entities are as follows:

1) TA: TA is a trusted authority whose communication area forms an independent trust domain. It is responsible for initializing the system and generating the public and private keys for certain entities. Furthermore, the true identity of the vehicles can only be revealed by the TA.

2) TEC: TEC is a trusted edge center with a communication area in the edge domain. It is responsible for the authentication and access of vehicle platoons within its communication range. Additionally, it acts as a node in the blockchain, enabling transaction retrieval and smart contract invocation.

3) RSU: RSU is a semi-trusted entity, assumed to be honest but curious. Deployed roadside, it connects securely to TEC via wired links and wirelessly to vehicles, acting as a communication bridge between the platoons and the TEC.

4) VP: VP comprises a Platoon Leader (PL) and multiple Platoon Followers (PF). The PL is a trusted entity, typically designated by an RSU or cloud server, elected through a capability-based voting mechanism among vehicles, or preassigned by human operators prior to platoon formation. The PL is responsible for managing internal coordination and handling communication with external entities. PFs are considered semi-trusted entities, meaning that they are generally

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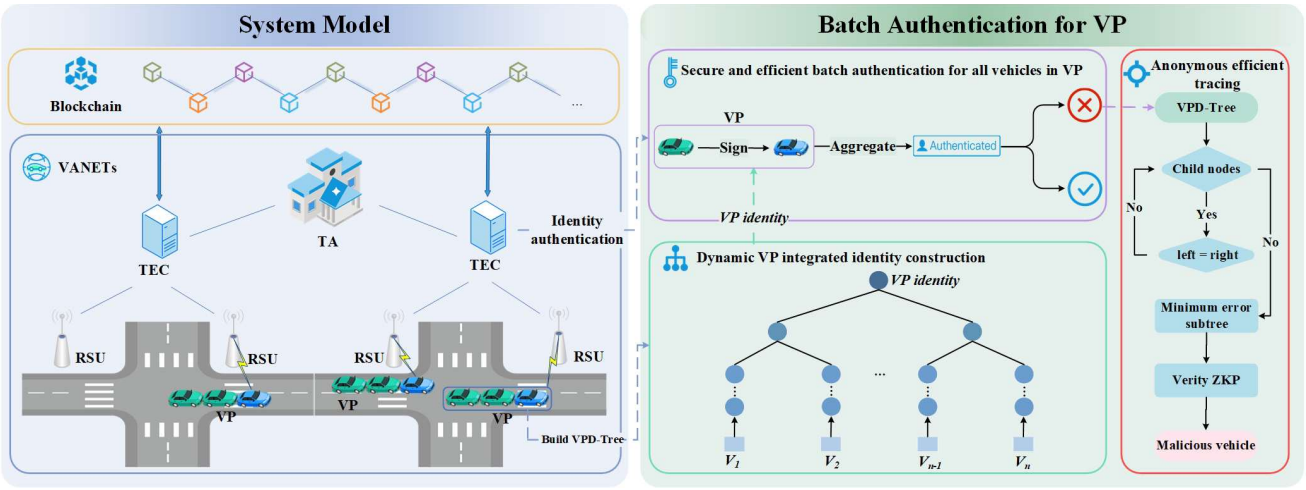


Fig. 1. System architecture of batch authentication for vehicle platoon in VANETs. The system model is composed of VANET infrastructure and blockchain. The batch authentication process includes dynamic VP identity construction, hash tree-based structure generation, and aggregated signature verification. Upon authentication failure, an anonymous efficient tracing mechanism is invoked to locate the malicious vehicle.

compliant but may occasionally exhibit misbehavior. During system operations, they are expected to follow the instructions issued by the PL.

5) BC: BC is a consortium blockchain maintained by TEC, storing vehicle identity authentication data, such as pseudonyms. This data is recorded as transactions, accessible to all nodes. The blockchain supports user registration, information queries, and smart contract integration for automated execution, reducing transaction delays.

Based on the infrastructure, we have designed the main process of the system to ensure secure and efficient batch authentication for VP in VANETs. The process is divided into the following three steps.

1) Dynamic VP Integrated Identity Construction: This is the foundation of VP identity authentication. First, TA performs system initialization and generate the necessary parameters and keys for the entire system. Following this, each vehicle in VP dynamically generates a pseudonym and computes the corresponding signature. The PF sends the relevant vehicle information to the PL, and the PL constructs the VP integrated identity by designing the VPD-Tree, preventing identity leakage, thereby protecting vehicle privacy.

2) Secure and Efficient Batch Authentication for All Vehicles in VP: After constructing the VP integrated identity information, the signatures of all vehicles in VP are aggregated to generate an aggregate signature for batch authentication. This aggregate signature is then sent through the RSU to the TEC for verification. During this process, quantum-resistant encryption algorithms are employed to ensure that the system remains secure in the era of quantum computing. This approach not only effectively reduces the transmission and storage overhead of signatures but also significantly improves the efficiency of batch authentication.

3) Anonymous Efficient Tracing: When VP identity authentication fails, the PL utilizes the previously constructed VPD-Tree to quickly trace potential malicious vehicles. Through binary search, the system can rapidly locate suspicious malicious vehicles and use ZKP to further confirm the identity of the malicious vehicle. This process enables conditional traceability of malicious vehicles by authorized entities, while ensuring that the privacy of legitimate participants remains protected. It achieves an efficient tracing mechanism that discloses only the necessary identity information of malicious vehicles, without compromising the anonymity of legitimate participants.

B. Threat Model

According to the Dolev-Yao threat model, the system's threat model is defined as follows:

1) The attacker can access the data on the blockchain but cannot tamper with it. However, they can observe and attempt to exploit the public information to analyze the vehicle's historical identity, thereby compromising anonymity.

2) The attacker possesses quantum computing capabilities, meaning they can use quantum computing to break traditional encryption algorithms that are not resistant to quantum attacks.

3) The attacker may attempt to forge the identity of a legitimate vehicle to join the platoon and falsify signatures for authentication.

4) The attacker may replay historical messages, attempting to use previously authenticated information or pseudonyms to masquerade as a legitimate vehicle and rejoin the platoon or communicate with the RSU.

5) PF and RSU are semi-trusted entities in the system. These entities are generally considered honest but curious, and they may exhibit potential malicious behavior or be subject to attacks, meaning they cannot be fully trusted.

6) The protocol allows for some participants to be malicious. Malicious participants may attempt to tamper with messages, leak keys, or collaborate with attackers to carry out replay and forgery attacks.

C. Security Requirements

The proposed scheme must satisfy the following security requirements.

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1) Privacy Preserving: The openness and dynamism of VP risk exposing vehicle identities during communication. To mitigate this, real identities must not be revealed during authentication, preventing attackers from deducing them through intercepted messages.

2) Unlinkability: Attackers may attempt to infer private information by collecting and analyzing large amounts of messages. Therefore, it is crucial to prevent an adversary from linking any two messages from the same vehicle.

3) Traceability: When malicious behavior occurs, effective scheme must be established to address identity tracking issues and prevent vehicles from exploiting privacy protection attributes. This property allows trusted authorities to trace the real identity of malicious vehicles.

4) Non-Repudiation of Signatures: The legitimate signature of a signer can prevent malicious adversaries from discovering their keys. However, without valuable information, malicious adversaries cannot generate valid aggregate signatures.

5) Quantum Resistance: This property ensures that the system remains resistant to quantum computer-based attacks, thus safeguarding the security of the system's signatures, encryption, and communication.

6) Resistance to Other Attacks: In addition to the above properties, secure identity authentication in VP should resist replay attacks and man-in-the-middle attacks. With the application of blockchain, the protocol should also be resistant to birthday collisions and hijacking attacks.

IV. Proposed Scheme

In this section, we will provide a detailed description of the specific process of the LETA scheme. The scheme consists of four steps: (1) initialization phase, where where the TA sets the global system parameters during the initial deployment of the vehicular network to support the subsequent authentication and tracing processes. (2) VP integrated identity construction phase, where a vehicle requesting to join a platoon registers with the TA to obtain a private key and a pseudonym. The vehicle then generates a signature and sends a request to PL, who constructs a VPD-Tree and builds the VP-integrated identity expression for the platoon. (3) VP identity authentication phase, where PL collects all requests and generates an aggregate signature to perform batch authentication of the VP identities efficiently. (4) Malicious vehicle tracing phase, where the PL initiates a VPD-tree-based tracing process upon detecting a failure in aggregate signature verification, enabling efficient identification of the malicious vehicle without compromising the anonymity of legitimate participants. The following will provide a detailed explanation of the proposed scheme. For better understanding, we have included the main notations and their descriptions, as shown in Table I.

TABLE I
Main Notations and Description

Notation	Description
$h(.)$	One-way collision hash function
R	Set of real numbers
q	Real number
$\mathcal{D}_{R,s}$	Discrete gaussian distribution
$\mathbb{Z}_q$	The ring of integers modulo q
$(A,b),s$	Master public key and master secret key
R_{V_i}, R_{RSU_i}	Real identities of Vehicle and RSU
P_{V_i}, P_{RSU_i}	Vehicle and RSU pseudo identities
s_{V_i}, s_{RSU_i}	Vehicle and RSU private key
$(A,b_{v_i}),(A,b_{RSU_i})$	Vehicle and RSU public key
Z_{V_i}	Zero-knowledge proof for vehicles
z	Random vector
μ_i	Commitment value
r	Response
Agg_{Sign}	Aggregate signature of vehicle formations

A. Initialization Phase

First, TA selects a security parameter λ determine the security of the lattice-based cryptosystem. A large prime number q then chosen as the modulus for the R-LWE problem. Next, a polynomial ring $R_q = \mathbb{Z}_q[x]/\langle x^n+1\rangle$ is defined, and $A \in R_q^{k\times l}$ is selected, where A is an element of the ring. Then, $s_1, s_2 \in D_{R,s}$ are chosen, where s_1 and s_2 are small polynomial vectors of dimensions $k\times 1$ and $l\times 1$, respectively. These form the secret key $s=(s_1,s_2)$, and the public key component is computed as $b = A\cdot s_1 + s_2 \mod q$, generating the public key pair (A,b). To further enhance the security of the protocol, three one-way collision-resistant hash functions are selected.

$$h_0 : \{0,1\}^* \to R_q \tag{1}$$

$$h_1 : \{0,1\}^* \to \{v : v \in R_q, \|v\| \le \epsilon_1\} \tag{2}$$

$$h_2 : \{0,1\}^* \to B_\epsilon^*, \quad B_\epsilon = \{p \in R : \|p\| \le \epsilon_2\} \tag{3}$$

Then, TA securely maintains the main secret key s and publicly sets the parameters as params $= \{k, q, h_0, h_1, h_2, A, b\}$.

B. VP Integrated Identity Construction Phase

1) Vehicle Registration: The vehicle V_i generates its real identity identifier $R_{V_i} = h_0(ID_{V_i} \parallel N_{V_i} \parallel s)$ based on the real identity ID_{V_i}, the driver's name N_{V_i}, and the private key s, and submits it to the Trusted Authority (TA). Then, V_i submits R_{V_i} to the TA. The TA randomly selects a number $\theta_{V_i} \in D_{R,s}$ and creates a pseudonymous identity $P_{V_i} = R_{V_i} \oplus h_0(\theta_{V_i})$ based on the R_{V_i} submitted by V_i.

Meanwhile, V_i generates its private key $s_{V_i} = (s_1, s_2)$ using the discrete Gaussian distribution, where each polynomial vector in the private key pair s_1 and s_2 is sampled from the discrete Gaussian distribution:

$$s_1 = (s_{1,1}, s_{1,2}, \ldots, s_{1,n}), \quad s_{1,i} \in D(R,s) \tag{4}$$

$$s_2 = (s_{2,1}, s_{2,2}, \ldots, s_{2,n}), \quad s_{2,i} \in D(R,s) \tag{5}$$

The private key $s_{V_i} = (s_1, s_2)$ is generated, and then the public key (A, b_{V_i}) is computed by $b_{V_i} = A\cdot s_1 + s_2 \mod q$. V_i submits the public key (A, b_{V_i}) to the TA, which saves the public key and binds P_{V_i} to the public key. Finally, $\{P_{V_i}, (A, b_{V_i}), T_1\}$ is uploaded to the blockchain for storage, where T_1 is the current timestamp.

2) RSU Registration: The RSU_i registration is similar to that of vehicles, generating the corresponding pseudonymous identity P_{RSU_i}, private key $s_{RSU_i} = (s_1, s_2)$ and public key

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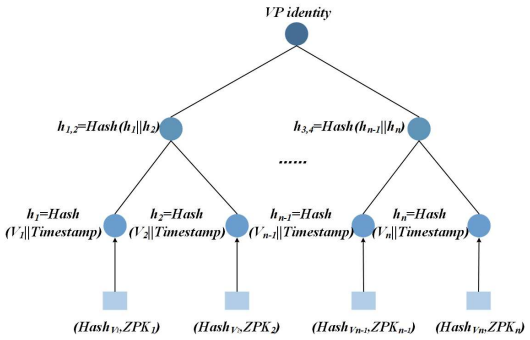


Fig. 2. VPD-tree.

(A, b_{RSU_i}). Finally, $\{P_{RSU_i}, (A, b_{RSU_i}), T_1\}$ is uploaded to the blockchain for storage.

3) Signature Generation Phase: After registration, the vehicle V_i generates a signature for the message $m_i \in \{0,1\}^*$ based on the input pseudonymous identity P_{V_i}, private key s_{V_i}, and public key (A, b_{V_i}), resulting in $Sign_{V_i}$. The specific process is as follows:

a) The vehicle V_i generates two random vectors $z_{i,1}, z_{i,2} \in (\mathbb{Z}_q[x]/\langle x^n + 1\rangle)^l$, ensuring that the coefficients of each polynomial are small, and then calculates the temporary commitment value μ_i:

$$\mu_i = A \cdot z_{i,1} + z_{i,2} \mod q. \tag{6}$$

b) The vehicle processes the commitment value μ_i and the message m_i using the hash function h_2, generating the challenge value c_i for the signature:

$$c_i = h_2(b_{V_i} \parallel \mu_i \parallel m_i) \tag{7}$$

c) The vehicle uses the private key s_{V_i} to calculate the signature response values $r_{i,1} = z_{i,1} + c_i \cdot s_1 \mod q$ and $r_{i,2} = z_{i,2} + c_i \cdot s_2 \mod q$, and then generates the final response value r_i:

$$r_i = r_{i,1} + A^{-1} \cdot r_{i,2} \mod q \tag{8}$$

d) The vehicle uses the commitment value μ_i, the response r_i, and the challenge c_i as components of a zero-knowledge proof, generating the zero-knowledge proof $Z_{V_i} = (\mu_i, r_i, c_i)$ to prove that the signature was generated using the private key s_{V_i}.

e) The vehicle V_i generates the signature $Sign_{V_i} = (r_i, c_i)$.

f) The vehicle V_i uploads $\{P_{V_i}, h_0(m_i), (A, b_{V_i}), r_i, Z_{V_i}, T_2\}$ to the blockchain, where T_2 is the current timestamp.

4) VPD-Tree: The VP is composed of a PL and multiple PF, with subsequent vehicles joining the vehicular network in the form of a convoy. The PL is responsible for guiding the convoy's route and speed, while the PF adjusts their driving paths and speeds based on the instructions from the leading vehicle, thus maintaining the overall coordination of the convoy. The process for constructing the convoy identity is as follows:

a) Each vehicle in the convoy calculates its identity hash value $Hash_{V_i} = h_1(P_{V_i} \parallel r_i \parallel \eta_i \parallel T_2)$, where $\eta_i \in D_{R,s}$, based on the information stored in its OBU. Then, the PF sends $(Hash_{V_i}, Z_{V_i})$ to the PL. The PL uses $(Hash_{V_i}, Z_{V_i})$ as leaf nodes to construct a dynamic hash tree, VPD-Tree, illustrated in Fig. 2, with the detailed process described below.

- Each leaf node contains $Hash_{V_i}$, which represents the integrity and identity information of vehicle V_i. After this step, a set of leaf node hash values is obtained, denoted as:
$$\{Hash_{V_1}, Hash_{V_2}, \ldots, Hash_{V_n}\}$$
- The leaf nodes are paired up, and if there is an odd number of nodes, empty nodes are added to pair them. The hash value of each parent node is generated by combining the hash values of its two child nodes. For example, for leaf node hashes $Hash_{left}$ and $Hash_{right}$, the parent node hash is:
$$Hash_{parent} = h_1(Hash_{left} \parallel Hash_{right}) \tag{9}$$
- The process is iterated to generate intermediate nodes, continuing until only one top-level node remains, the root node.
- The hash value of the root node, *Root*, serves as the unique representation of the entire hierarchical hash tree, ensuring the integrity of all leaf nodes. The *Root* is then broadcast as the $VP\ identity$ to all vehicles in the platoon. When vehicles join or leave, only the partial update of relevant values is needed to ensure the efficient update of VP identity.
- When multiple vehicles request to join or leave the platoon simultaneously, the corresponding branches of the VPD-Tree can be updated independently and concurrently, rather than triggering a sequential update of the entire structure. This is achieved by maintaining localized subtree operations, which allow multiple update threads or processes to perform in parallel without conflicts.

b) V_i selects a random value $\delta_i \in R$, calculates $M_{i,1} = h_0(\delta_i)$, $M_{i,2} = h_0(M_{i,1}||T_3)$, and sends $\{P_{V_i}, M_{i,1}, M_{i,2}, T_3\}$ to PL, where T_3 is the current timestamp.

c) After receiving the information from all vehicles, PL recalculates $M'_{i,2} = h_0(M_{i,1}, T_3)$. If $M'_{i,2} = M_{i,2}$, then the vehicle's P_{V_i} is valid and the vehicle's registered identity is trusted.

d) Each vehicle in the formation negotiates and generates a session key for subsequent internal encrypted communication within the formation. PL first calculates the public value $B_{PL} = A \cdot s_1 \mod q$ and broadcasts it to all PFs. PF calculates the public value $B_i = A \cdot s_{1,i} \mod q$ based on its private key, and computes the session key $K_i = B_{PL} \cdot s_{1,i} \mod q$. PL calculates the session key $K_{session} = B_i \cdot s_1 \mod q$. Since both parties use the same module lattice and computation methods, it ensures the consistency of all session keys, i.e., $K_{session} = K_i$.

C. VP Identity Authentication Phase

1) Signature Aggregation: The PFs in the VP send their signatures to the PL, who then aggregates the signatures of the entire platoon. The specific aggregation process is as follows:

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a) Each vehicle sends its signature $\text{Sign}(V_i)$ to the PL.

b) PL calculates the hash value β_i for each vehicle:

$$h_2(\alpha_{(V_1)}, t_1, \mu_1, \ldots, \alpha_{(V_n)}, t_n, \mu_n, \text{VP identity}) \rightarrow (\beta_1, \beta_2, \ldots, \beta_n) \tag{10}$$

c) PL aggregates the challenge values c_i from all vehicles $(V_1, V_2, V_3, \ldots, V_n)$ to generate c_{agg}:

$$c_{\text{agg}} = \sum_{i=1}^{n} \beta_i \cdot c_i \mod q \tag{11}$$

d) The responses r_i from all vehicles $(V_1, V_2, V_3, \ldots, V_n)$ are aggregated to generate r_{agg}:

$$r_{\text{agg}} = \sum_{i=1}^{n} \beta_i \cdot r_i \mod q \tag{12}$$

e) Generate the aggregated signature $\text{Agg}_{\text{Sign}} = (r_{\text{agg}}, c_{\text{agg}})$.

f) The VP uses RSU_i as a relay device to send $\{\text{VP identity}, \text{Agg}_{\text{Sign}}, T_5\}$ to the corresponding TEC, where T_5 is the timestamp when the aggregated signature is generated.

2) Signature Verification: After receiving the aggregated signature Agg_{Sign} sent by the RSU, the TEC will verify the signature. The specific verification process is as follows:

a) The RSU_i first verifies the timestamp T_5. If $T - T_5 > \Delta T$, it is considered a malicious signature, and the RSU will reject the message. If T_5 is within the allowed time range, the message is considered valid, and the subsequent signature verification can proceed.

b) Recalculate each vehicle's hash value β_i according to Eq. 10.

c) Verify whether the following equation holds:

$$A \cdot r_{\text{agg}} = \sum_{i=1}^{n} \beta_i \cdot (\mu_i + c_i \cdot b_{V_i}) \mod q \tag{13}$$

If Eq. 13 does not hold, it indicates the presence of malicious vehicles in the convoy. If Eq. 13 holds, the verification process accepts the correctness of the aggregated signature.

D. Malicious Vehicle Tracing Phase

When aggregated signature verification fails, it indicates the presence of malicious vehicles in the platoon, requiring a trace-back mechanism to identify and revoke their identities. Following Algorithm 1, the VPD-Tree employs a hierarchical structure that organizes vehicles across multiple levels, where the root node represents the entire platoon and the leaf nodes correspond to individual vehicles. The malicious vehicle tracing mechanism leverages this structure to efficiently locate suspected vehicles and utilizes ZKP embedded in the leaf nodes to verify identities without disclosing confidential information, thereby enabling anonymous and efficient tracing within the VP.

1) Starting from the root node, dynamically calculate the hash value as $Root' = h_1(Hash_{left}||Hash_{right})$. If the calculated $Root'$ is not equal to the original *Root*, proceed to the child nodes for further examination.

Algorithm 1 Efficient Malicious Vehicles Tracing

Input: Vehicle platoon information based on $VPD-Tree$
Output: ZKP for malicious vehicles

```
for VPD-Tree do
  if The children of the Root is Null and verity ZKP = False then
    return ZKP;
  else
    Root' = h1(Hash_left || Hash_right);
    if Root' != Root then
      if Children'_left != Children_left then
        error_Tree = error(Children_left);
      end if
      if Children'_right != Children_right then
        error_Tree = error(Children_right);
      end if
    end if
    return leaf nodes;
  end if
end for
Find the suspicious leaf nodes;
Verify the ZKP of the suspicious leaf nodes;
if VerifyZKP(LeafNode.ZKP) == False then
  Mark the leaf as malicious;
end if
return the ZKP for malicious vehicles;
```

2) For each child node, dynamically recalculate its hash value and compare $Hash'$ with *Hash*. If the values do not match, continue to examine its child nodes. If they match, it indicates that this subtree is valid, and that branch can be excluded from further investigation.
3) Recursively trace down to the leaf node layer, ultimately locating the smallest problematic subtree. This tracing process can be performed in parallel across disjoint branches, significantly improving efficiency in large-scale platoon environments.
4) After locating the smallest problematic subtree, the authenticity of the node's data is verified through the zero-knowledge proof $Z_{V_i} = (\mu_i, r_i, c_i)$. In the context of the VPD-Tree, each leaf node stores the ZKP of the corresponding vehicle, enabling privacy-preserving identity verification without exposing sensitive identity information. First, the commitment value μ_i' is recalculated as:

$$\mu_i' = A \cdot r_i - c_i \cdot b_{V_i} \mod q \tag{14}$$

Next, the consistency of the commitment value is checked. If $\mu_i \neq \mu_i'$, the node is immediately flagged as a potential malicious node. If $\mu_i = \mu_i'$, the challenge value c_i' is calculated as:

$$c_i' = h_2(b_{V_i} \parallel \mu_i \parallel m) \tag{15}$$

If $c_i = c_i'$, the node is considered legitimate; otherwise, it is marked as a malicious node.
5) The PL identifies the pseudonym of the malicious vehicle through the blockchain and submits it to the TA

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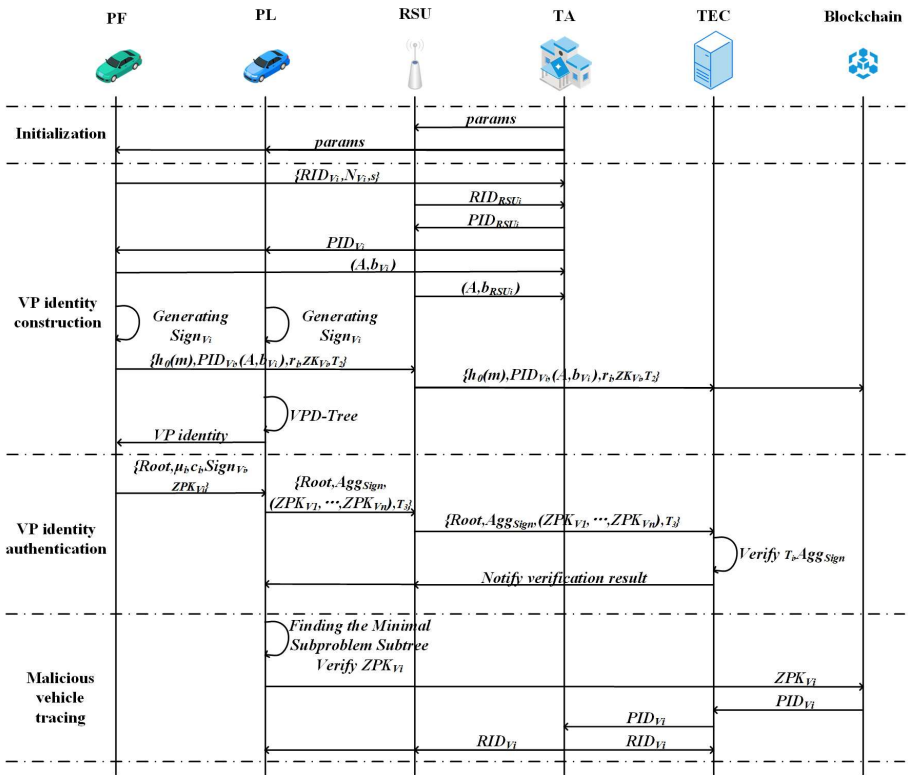


Fig. 3. Overall process workflow of LETA.

without accessing the vehicle's real identity. The TA, as the only entity capable of resolving pseudonyms to real identities, performs the identity revocation, updates the platoon records, and securely broadcasts the necessary changes to nearby vehicles and RSUs.

6) To prevent identity link attacks and protect vehicle privacy, the TA will update the pseudonyms of vehicles and RSUs after a certain period of time. The TA will re-select random numbers θ'_{V_i} and $\theta'_{\mathrm{RSU}_i} \in D_{R,s}$, and regenerate the pseudonyms as follows:

$$\mathrm{P}'_{V_i} = \mathrm{R}_{V_i} \oplus h_0(\theta'_{V_i}), \tag{16}$$

$$P'_{RSU_i} = R_{RSU_i} \oplus h_0(\theta'_{RSU_i}). \tag{17}$$

The TA will issue new pseudonyms to vehicles and RSUs for identity updates. The smart contract will replace the original pseudonyms P'_{V_i} and P'_{RSU_i} on the blockchain The PL will update the relevant leaf nodes in the VPD-Tree.

The overall flow diagram of our proposed framework is illustrated in Fig.3.

V. Security Analysis

A. Proof of Correctness

In this section, we will discuss the correctness proof of the final aggregated signature and verify the reliability of the aggregated signature Agg_{Sign}. The correctness of the aggregated signature is shown as follows:

$$\begin{aligned}
&\mathrm{A} \cdot r_{\mathrm{agg}} \\
&= \mathrm{A} \cdot \sum_{i=1}^{n} \beta_i \cdot r_i \bmod q \\
&= \mathrm{A} \cdot \sum_{i=1}^{n} \beta_i \left(r_{i,1} + \mathrm{A}^{-1} r_{i,2}\right) \bmod q \\
&= \mathrm{A} \cdot \sum_{i=1}^{n} \left(\beta_i \left(z_{i,1} + c_i \cdot s_1\right) + \mathrm{A}^{-1} \beta_i \left(z_{i,2} + c_i \cdot s_2\right)\right) \bmod q \\
&= \mathrm{A} \cdot \sum_{i=1}^{n} \beta_i (z_{i,1} + c_i \cdot s_1) + \sum_{i=1}^{n} \beta_i (z_{i,2} + c_i \cdot s_2) \bmod q \\
&= \sum_{i=1}^{n} \beta_i (\mathrm{A} \cdot z_{i,1} + \mathrm{A} \cdot c_i \cdot s_1 + z_{i,2} + c_i \cdot s_2) \bmod q \\
&= \sum_{i=1}^{n} \beta_i \cdot (\mu_i + c_i \cdot (\mathrm{A} \cdot s_1 + s_2)) \bmod q \\
&= \sum_{i=1}^{n} \beta_i \cdot (\mu_i + c_i \cdot b_{V_i}) \bmod q
\end{aligned}$$

B. Formal Analysis

In this chapter, we analyze the unforgeability of the aggregated signature under the Q-Random Oracle Model (Q-ROM) [34], demonstrating that attackers cannot forge a valid signature in the proposed protocol. Q-ROM, an extension of the traditional Random Oracle Model (ROM), accommodates quantum environments by enabling quantum parallel queries. The protocol's security, based on the R-LWE assumption, ensures resistance to quantum attacks. Using the game sequence technique [35], we formally prove the protocol's post-quantum security by modeling interactions between quantum polynomial-time adversaries and challengers through a response-challenge game. The key components of Q-ROM are outlined below.

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Definition 1 (Participants and Partnering): Participants consist of TA, TEC, PL, and PF. In the i-th instance, the participants are represented as $\Psi^i_{TA_i}$, $\Psi^i_{TEC_i}$, $\Psi^i_{PL_i}$, and $\Psi^i_{PF_i}$, respectively. During the protocol execution, TEC, PL, and PF share the same global public key A and use a consistent signature verification mechanism to validate the legitimacy of signatures. Thus, TEC, PL, and PF form a partnership.

Definition 2 (Queries): Queries simulate the capabilities of the attacker.

Execute ($\Psi^i_{TA_i}, \Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}$): The attacker passively records all interactions among vehicles, RSUs, and TEC during protocol execution.

Send ($\Psi^i_{TA_i}, \Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}, Aggsign$): A forges and sends a counterfeit aggregated signature $Aggsign^*$ to $\Psi^i_{TEC_i}$. If $Aggsign^*$ is successfully verified, $\Psi^i_{TEC_i}$ responds to A.

Reveal ($\Psi^i_{TA_i}, \Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}$): This query simulates key leakage to the attacker. If the correct aggregated signature cannot be generated, it indicates no key leakage.

Corrupt ($\Psi^i_{PL_i}, \Psi^i_{PF_i}$): This query simulates the attacker gaining full control over a vehicle's internal state and accessing all its stored information.

Test ($\Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}$): This query simulates an adversary's attack on the protocol's unforgeability. The attacker selects a signature, and the challenger randomly returns either a genuine or forged aggregated signature, which the attacker must distinguish.

Definition 3 (Freshness): An instance is considered fresh if it satisfies the following conditions: 1) The submitted aggregated signature is confirmed to have been computed by the instance $\Psi^i_{PL_i}$; 2) The attacker has not performed a Reveal query on $\Psi^i_{PL_i}$ or its partners. If these conditions are met, the instance is deemed fresh.

Definition 4 (Semantic Security): Within quantum polynomial time, the attacker is allowed to perform at most one Test($\Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}$) query and multiple other queries to determine the correctness of the return value from Test($\Psi^i_{TEC_i}, \Psi^i_{PL_i}, \Psi^i_{PF_i}$). Specifically, the attacker guesses the aggregated signature in the Test query and outputs a guessed value. The advantage of the attacker is defined as:

$$\mathrm{Adv}_P^{\mathcal{A}} = |2 \cdot \Pr[\mathrm{suc}(\mathcal{A})] - 1|. \tag{18}$$

where $\mathrm{Adv}_P^{\mathcal{A}} < \eta$ indicates that the protocol is secure, and η is sufficiently small.

Lemma 1 (Difference Lemma [36]): Let X, Y, Z represent events in a probability space, and suppose the event $Y \wedge \neg Z$ is equivalent to the event $X \wedge \neg Z$. Then, $\Pr[X] - \Pr[Y] \leq \Pr[Z]$.

Theorem 1: In the Q-ROM model, define $Adv_{\mathcal{A}}$ as the probability that an adversary $\mathcal{A}_1$ forges a valid aggregated signature within a finite time t. Suppose the adversary $\mathcal{A}$ makes q_0 queries to h_0, q_1 queries to h_1, q_2 queries to h_2, q_{sk} private key s queries, and q_s signing *Sign* queries. The advantage of the adversary $\mathcal{A}$ in forging an aggregated signature is bounded by:

$$Adv_{\mathcal{A}} \leq \frac{q_0^3 + q_1^3}{2^n} + \frac{2\sqrt{q_2}}{2^{\lambda/2}} + \frac{q_s + q_{sk}}{2^{\lambda-1}} + 2Adv_{\mathcal{A}}^{R-LWE} \tag{19}$$

where $Adv_{\mathcal{A}}^{R-LWE}$ is the probability of solving the R-LWE problem.

Proof: Suppose adversary $\mathcal{A}$ can forge the aggregated signature. In this case, challenger $\mathcal{B}$ can leverage the query behavior of adversary $\mathcal{A}$ to extract the solution to the R-LWE problem. A series of security games is used to complete the formal security proof, where Win_i denotes challenger $\mathcal{B}$'s success in completing $Game_i$.

Game$_0$: Adversary $\mathcal{A}$ fully simulates the real behavior of the signature system without any additional interference. Based on the definition of semantic security, we can derive:

$$Adv_{\mathcal{A}} = |2 \cdot \Pr[Win_0] - 1|. \tag{20}$$

Game$_1$: Challenger $\mathcal{B}$ replaces the hash functions h_0, h_1, and h_2 with random oracles (RO_1, RO_2, RO_3) and simulates the behavior of these oracles. In the Q-ROM, since adversary $\mathcal{A}$ cannot distinguish between the real hash functions and the random oracles, it cannot differentiate this game from the actual execution of the protocol. Therefore:

$$|\Pr[Win_1] - \Pr[Win_0]| = 0. \tag{21}$$

Game$_2$: Similar to *Game$_1$*, h_0 and h_1 queries are simulated. If adversary $\mathcal{A}$'s queries to h_0 or h_1 result in a collision (i.e., two different inputs produce the same hash output), the game terminates immediately. Based on the birthday paradox, we can derive:

$$|\Pr[Win_2] - \Pr[Win_1]| \leq \frac{q_0^3}{2^{n+1}} + \frac{q_1^3}{2^{n+1}}. \tag{22}$$

Game$_3$: Adversary $\mathcal{A}$ forges a valid signature $Sign_{V_i} = (r_i, c_i)$ without relying on the private key s_{V_i}. The adversary uses queries to h_2 to forge a specific challenge value c_i that satisfies the signature verification condition. After obtaining μ_i and c_i, the attacker attempts to forge the response value r_i to make the verification equation hold.

To succeed, the attacker must find a valid c_i through h_2 queries. Given that the adversary can perform q_2 queries, the maximum success probability for forging is $\sqrt{q_2}/2^{\lambda/2}$. Therefore, we have:

$$|\Pr[Win_3] - \Pr[Win_2]| \leq \frac{\sqrt{q_2}}{2^{\lambda/2}}. \tag{23}$$

Game$_4$: Challenger $\mathcal{B}$ simulates the process of generating the aggregated signature. Adversary $\mathcal{A}$ can make up to q_s signing queries on messages $\{m_1, m_2, \ldots, m_n\}$ and attempts to forge a valid aggregated signature, where at least one individual signature $Sign_i$ was not included in the signing queries. The maximum success probability for adversary $\mathcal{A}$ forging a correct aggregated signature is $q_s/2^{\lambda}$. Therefore, we have:

$$|\Pr[Win_4] - \Pr[Win_3]| \leq \frac{q_s}{2^{\lambda}}. \tag{24}$$

Game$_5$: Challenger $\mathcal{B}$ replaces the original private key $s_{V_i} = (s_1, s_2)$ with a random key $s^*_{V_i} = (s'_1, s'_2)$. Based on the R-LWE assumption, $s^*_{V_i} = (s'_1, s'_2)$ is computationally indistinguishable from $s_{V_i} = (s_1, s_2)$. During the signature generation simulation, all computations involving the original private key are replaced with computations based on the random key $s^*_{V_i}$.

If adversary $\mathcal{A}$ attempts to approximate the real private key through private key queries, the maximum probability of success after q_{sk} queries is $q_{sk}/2^{\lambda}$. On the other hand, if adversary

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TABLE II
COMPARISON OF SECURITY FEATURES

Scheme	[37]	[38]	[39]	[30]	LETA
Man-in-the-Middle	✓	✓	✓	✓	✓
Replay Attack	✓	✓	✗	✗	✓
Forgery Attack	✓	✗	✓	✓	✓
Impersonation attack	✗	✓	✗	✗	✓
Conditional privacy protection	✓	✓	✓	✓	✓
Unlinkability	✗	✗	✗	✓	✓
Anonymous traceability	✗	✗	✗	✓	✓
Authentication and integrity	✓	✓	✓	✓	✓
Quantum attack resistance	✗	✗	✓	✓	✓

A successfully computes $s_{V_i} = (s_1, s_2)$,where $s_2 = b - As_1$, the success probability is bounded by Adv_A^{R-LWE}, which is the advantage of solving the R-LWE problem. Therefore, we have:

$$|\Pr[Win_5] - \Pr[Win_4]| \leq Adv_{\mathcal{A}}^{R-LWE} + \frac{q_{sk}}{2^{\lambda}}. \tag{25}$$

In fact, if adversary $\mathcal{A}$ fails to make any random oracle queries with the correct input during the game, $\mathcal{A}$ gains no advantage in distinguishing the real private key s_{V_i} from a fake key of the same length created using random values. Therefore, $\Pr[Win_5] = 1/2$. We can further derive:

$$\begin{aligned} &Adv_{\mathcal{A}} \\ &= |2\Pr[Win_0] - 1| \\ &= 2\Pr[Win_5] - 1 + 2(\Pr[Win_0] - \Pr[Win_5]) \\ &\leq 2|\Pr[Win_0] - \Pr[Win_1]| + \cdots + 2|\Pr[Win_4] - \Pr[Win_5]| \\ &\leq \frac{q_0^3 + q_1^3}{2^n} + \frac{2\sqrt{q_2}}{2^{\lambda/2}} + \frac{q_s + q_{sk}}{2^{\lambda-1}} + 2Adv_{\mathcal{A}}^{R-LWE}. \end{aligned} \tag{26}$$

This result demonstrates that the aggregated signature protocol possesses unforgeability.

C. Other Security Analysis

We compared LETA with the schemes [30], [37], [38], and [39] to evaluate their security features. Scheme [37] cannot resist impersonation attacks or quantum attacks and fails to ensure unlinkability and anonymous traceability. Scheme [38] fails to provide anonymous traceability and quantum resistance. Scheme [39] cannot resist replay attacks and impersonation attacks and also lacks unlinkability and anonymous traceability. Scheme [30] fails to resist replay attacks and impersonation attacks. A detailed comparison of security features is shown in the table, where "✓" indicates that the scheme satisfies the corresponding security feature, and "✗" indicates that it does not. The results show that Table II, compared to other related schemes, the proposed authentication scheme offers more comprehensive support for security features. Therefore, LETA provides higher security and more reliable protection for VP.

Next, a detailed analysis of the security features in Table II will be presented.

1) Conditional Privacy Protection: To prevent unauthorized identity access, pseudonyms are generated for vehicles and RSUs in VANETs. P_{V_i} and P_{RSU_i} represent the pseudonyms of vehicles and RSUs, respectively. These pseudonyms are issued by the TA during their respective registration phases. The TA is the sole entity with access to true identities, ensuring the pseudonyms cannot reveal real identities.

2) Unlinkability: In the proposed scheme, vehicle pseudonyms are generated as $P_{V_i} = R_{V_i} \oplus h_0(\theta_{V_i})$, where h_0 is a one-way collision-resistant hash function. Pseudonyms are dynamic, and the platoon is constructed using a dynamic hash tree with random values to enhance unlinkability. Thus, no one can link pseudonyms or platoon data to a vehicle's real identity.

3) Message Authentication and Integrity: To ensure identity authentication and message integrity, each message is signed and aggregated by the registered vehicles before being sent to the TEC. The TEC can then verify the validity of the aggregated signature. A formal proof that the aggregated signature cannot be maliciously forged has been provided above.

4) Anonymous Traceability: The identities and signatures of platoon vehicles are mapped to the VPD-Tree. In the event of a platoon identity authentication failure, the VPD-Tree can be used to quickly locate the malicious vehicle. ZKP is then employed to verify the malicious vehicle, ensuring its identification while preserving the privacy of the VP.

5) Non-Repudiation: The unforgeability of the aggregated signature ensures that the VP cannot deny sending a message. Upon detecting malicious behavior, the verifier can link the signature to a specific VP and, using the VPD-Tree and the TA, identify the malicious vehicle. Thus, the protocol guarantees non-repudiation.

6) Resistance to Quantum Attacks: RSA is a public-key cryptosystem based on the integer factorization problem, while ECC is based on the discrete logarithm problem. However, these problems can be efficiently solved in a quantum computing environment. As a result, lattice-based cryptography has gained significant attention as a means to resist quantum attacks. The proposed aggregated signature protocol is built on the R-LWE problem, and its security relies on the computational complexity of this problem. Since quantum computers cannot efficiently solve the R-LWE problem, the protocol remains secure in a quantum computing environment.

7) Resistance to Other Security Attacks:

a) Man-in-the-middle attack: The VP performs identity authentication through signature verification, ensuring the legitimacy of each vehicle's identity within the platoon. Therefore, the proposed protocol is resistant to man-in-the-middle attacks.

b) Replay attack: Each message includes a timestamp to prevent replay attacks. The verifier first checks the timestamp to ensure the freshness of the message. Thus, the proposed protocol is resistant to replay attacks.

c) Forgery attack: The signature algorithm used in the protocol ensures message integrity. Any modification of the

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TABLE III
COMPARISON OF COMMUNICATION COST

Scheme	Single signature transmit(bytes)
SAS	$\|R_p\| \times \lceil log_2 P \rceil + L_h = 260$
LBAS	$\|R_p\| \times \lceil log_2 P \rceil + L_h = 260$
IBAGS	$\|R_p\| \times \lceil log_2 P \rceil = 240$
ES-DA-LBCLA	$\|R_q\| \times 2\lceil log_2 P \rceil + L_h = 340$
LETA	$\|R_q\| \times \lceil log_2 P \rceil + L_h = 180$

data will cause the verification equation to fail. Hence, the protocol is resistant to forgery attacks.

d) Impersonation attack: In the proposed protocol, correctness, unforgeability, and authenticity ensure that signatures generated using false identities cannot pass signature verification.

VI. PERFORMANCE ANALYSIS

To prove the efficiency of LETA, we conducted simulation experiments to evaluate the computational and communication cost of identity authentication. Additionally, we also analyze the efficiency of tracking malicious vehicles to highlight the performance of our tracking scheme. All experiments were implemented in Python and executed on a device running macOS with an M3 chip, featuring an 8-core CPU and a 10-core GPU. In the VP batch authentication experiments, the LETA scheme was compared with the following schemes:

SAS [39]: Wang and Wu designed an efficient lattice-based sequential aggregate signature scheme.

LBAS [29]: Li et al. proposed a lattice-based batch authentication scheme with dynamic identity revocation, which enhances verification efficiency and supports quantum-resistant security for VANETs.

IBAGS [30]: Prajapat et al. proposed a novel lattice-based identity-based aggregate signature system. This approach combines batch verification and aggregate signature mechanisms to authenticate multiple vehicle units and aggregate multiple user signatures into a single signature.

ES-DA-LBCLA [27]: Wani et al. proposed a lattice-based dynamic anonymous batch-verifiable authentication scheme for VANETs, which resists quantum attacks and eliminates pseudonym storage by using a fuzzy extractor for dynamic anonymity.

A. Analysis of Communication Cost

To evaluate the communication cost of our proposed scheme, the hash length(L_h) is set to 20 bytes, and q is set to 32 bytes. To ensure the same security level, R_q is set to 32 bytes and R_p is set to 48 bytes. Additionally, for comparison with other schemes, the parameters are configured with G_1 set to 128 bytes, G to 20 bytes, and Z_q^* to 40 bytes. In our proposed scheme, the signature consists of (r_i, c_i), so the message length sent to the TEC is $|R_q| \times \lceil log_2 P \rceil + L_h = 180$ bytes.

Similarly, we calculated the communication costs of the SAS, LBAs, IBAGS and ES-DA-LBCLA schemes, and the results are presented in Table III. Lattice-based cryptography uses higher-dimensional mathematical structures to resist quantum attacks, resulting in larger ciphertexts and signatures compared to traditional cryptography. Consequently, achieving quantum resistance comes with a slight increase in communication cost in our proposed protocol. However, because the modular lattice are used, LETA can provide stronger security with smaller parameters, so the communication cost of the proposed scheme is lower than that of other lattice-based schemes. To provide a more intuitive comparison of the differences between these schemes, we evaluated the communication cost under different numbers of vehicles, with the results shown in the Fig. 4. The analysis indicates that in single signature scenarios, our scheme achieves the lowest communication cost. Given that n increases, the communication cost of LETA remains the lowest compared to SAS, LBAs, IBAGS and ES-DA-LBCLA. Overall, compared to the aforementioned schemes, our proposed scheme consistently exhibits the lowest communication cost in all scenarios, highlighting its advantages in resource utilization and efficiency.

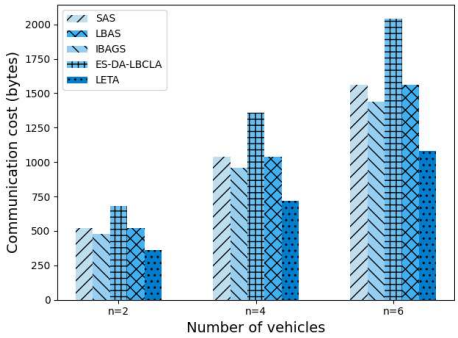


Fig. 4. Comparison of communication costs for different VP vehicle counts.

Overall, our scheme achieves lower communication costs compared to others, highlighting its efficiency and resource utilization, especially in VANETs. It minimizes data transmission while maintaining security, ensuring effective operation in resource-limited environments.

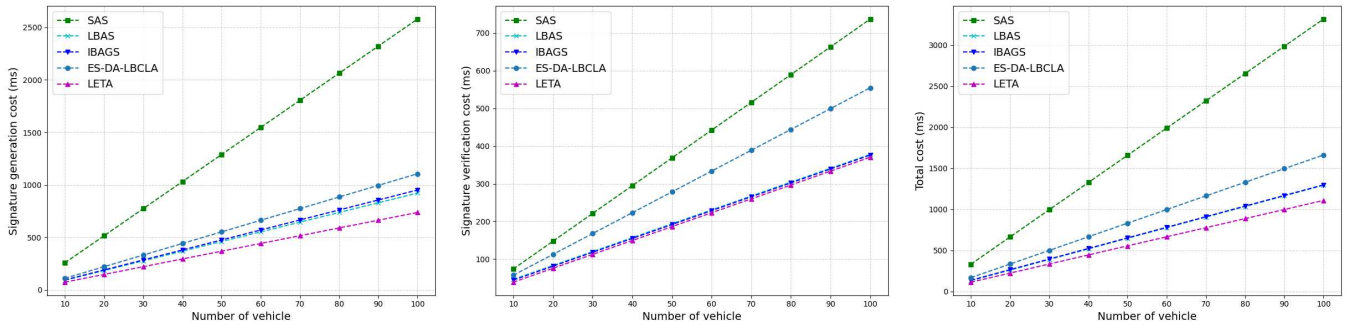
B. Analysis of Computation Cost

In this section, we present a detailed comparison of the computational costs of the proposed scheme with the existing aggregate signature schemes SAS, LBAs, IBAGS and ES-DA-LBCLA. The computational cost evaluation is based on the execution time of core cryptographic operations, including bilinear pairing, polynomial addition/multiplication, and hashing, as detailed in Table IV.

Our scheme includes detailed steps for signature generation and verification, with the execution time for each step listed in Table V. Specifically, the time required for generating a single signature is $4T_a + 3T_m + T_h \approx 5.5328ms$. During the signature verification phase, our scheme requires the least amount of time compared to other schemes, mainly due to the use of public key verification, which simplifies the relevant operations and reduces computational complexity.

Fig. 5(a) shows the computational cost of signature generation for different schemes, Fig. 5(b) shows the computational cost of signature verification for different schemes, and

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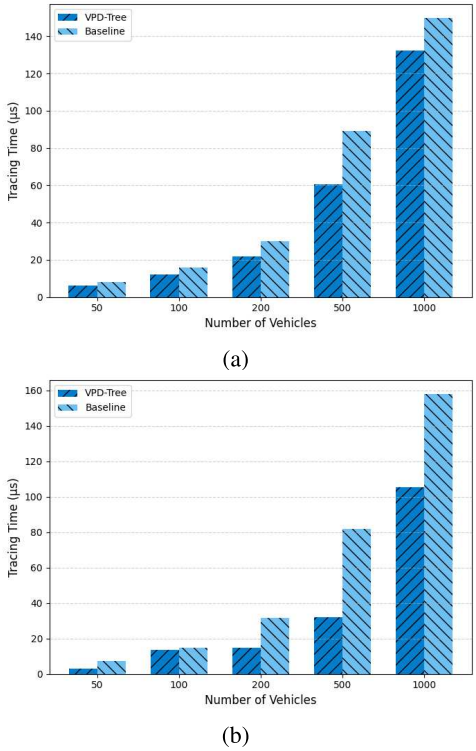
(a) signature generation. (b) signature verification. (c) Total computational cost.

Fig. 5. Comparison of computational cost at different stages.

TABLE IV
RELATED OPERATION TIME

Notation	Description	Operation time
T_{bp}	Bilinear pairing	3.3658ms
T_a	Polynomial addition	0.0014ms
T_m	Polynomial multiplication	1.8423ms
T_h	Hash operation	0.0003ms

Fig. 5(c) shows the total cost of identity authentication for these schemes. The results indicate that, compared to the SAS and ES-DA-LBCLA schemes, our scheme significantly reduces the computational cost at each step, demonstrating its advantage in minimizing resource consumption. Compared to the LBAS and IBAGS schemes, our solution maintains higher security and efficiency while slightly lowering computational costs. The improved computational efficiency can be largely ascribed to the fact that both the signature generation and verification procedures in the proposed scheme are based solely on lightweight polynomial addition and multiplication operations. These operations are inherently efficient and impose minimal computational burden, which contributes to reduced latency and lower resource consumption. The reduction in computational cost not only benefits real-time authentication but also helps conserve system resources, making the scheme particularly suitable for VANETs.



(a)

(b)

Fig. 6. (a) Tracing time with 5% malicious vehicles (b) Tracing time with 10% malicious vehicles.

C. Malicious Vehicle Tracing Efficiency Analysis

In this section, we evaluate the performance of the proposed scheme in tracing malicious vehicles. The time complexity of tracing using the VPD-Tree is $O(\log n)$, whereas the time complexity of directly traversing and verifying all vehicles is $O(n)$. This clearly demonstrates that our proposed method is significantly more efficient than traditional schemes, especially in large-scale scenarios.

To further validate this advantage, we conduct an ablation experiment comparing the tracing time of the VPD-Tree and the sequential re-authentication mechanism under different platoon sizes with 5% and 10% malicious vehicles, as shown in Fig. 6(a) and Fig. 6(b). The results show that the VPD-Tree consistently achieves lower tracing time, with the performance gap widening as the platoon size increases, confirming its superior scalability and efficiency in high-mobility VP environments.

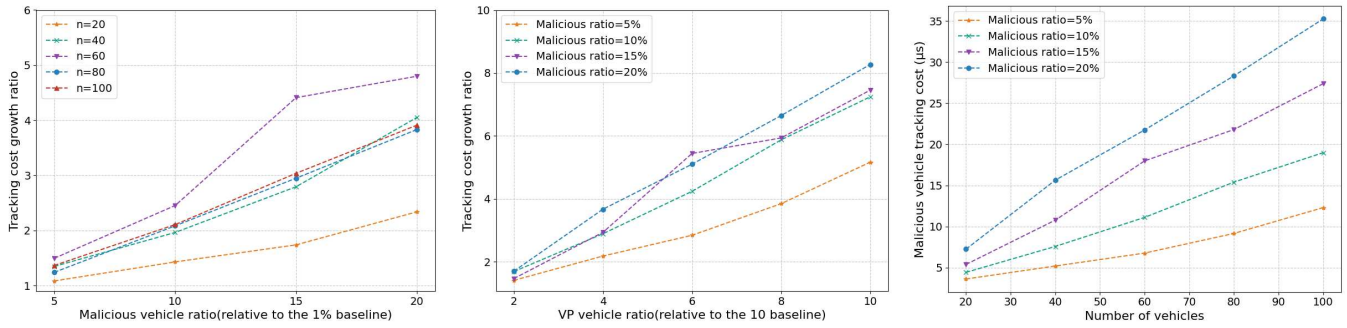
Specifically, in the experiments, the number of vehicles was set from 20 to 100, and the proportion of malicious vehicles was set at 5%, 10%, 15%, and 20%. To ensure the accuracy of the results, each scenario was tested five times, and the average was calculated. The Fig. 7 shows the malicious vehicle traceability based on VPD-Tree.

We increased the proportion of malicious vehicles step by step, using 1% as the baseline, to analyze the growth of malicious vehicle tracing cost, as shown in Fig. 7(a). The tracking cost growth ratio is calculated as the ratio of $time'/time$, where $time'$ is the time required for the comparison experiment, and *time* is the time required for the baseline. The results show that, in VP with different numbers

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TABLE V
COMPARISON OF COMPUTATION COST

Scheme	Signature	Aggregate signature	Verification
SAS	$4T_a + 12T_m + T_h$	$(n-1)T_a + 2nT_m$	$n(T_h + 2T_a) + 4nT_m$
LBAS	$4T_a + 4T_m + T_h$	$(3n-1)T_a + nT_m$	$3nT_a + (2n+1)T_m$
IBAGS	$4T_a + 4T_m + T_h$	$(n-1)T_a + nT_m$	$n(T_h + 2T_a) + (2n+1)T_m$
ES-DA-LBCLA	$4T_a + 4T_m + 2T_h$	$(3n-1)T_a + 2nT_m$	$2n(T_h + T_a) + (3n+1)T_m$
LETA	$4T_a + 3T_m + T_h$	$(2n-1)T_a + nT_m$	$(n-1)T_a + (2n+1)T_m$



(a) Malicious Vehicle Count Impact on Tracing Cost. (b) VP Vehicle Count Impact on Tracing Cost. (c) Tracing Cost for Different VP Vehicle Counts.

Fig. 7. Malicious vehicle tracing based on VPD-tree.

of vehicles, the increase in tracing costs is relatively small as the proportion of malicious vehicles increases. Even when the number of vehicles reaches 100, a 20-times increase in malicious vehicles leads to an increase in tracing cost of less than 5 times. Fig. 7(b) shows the trend of tracing cost as the number of vehicles increases, with 10 vehicles as the baseline. It can be seen that when the number of vehicles in the VP increases by a factor of 10, the traceability cost increases by about 8 times. This is mainly due to the increase in the number of layers in the VPD-Tree as the number of vehicles grows, leading to a significant increase in traceability cost, but the overall efficiency remains high. Fig. 7(c) shows that, even when there are 20% malicious vehicles in a VP of 100 vehicles, the traceability time is only about 35 μs, further substantiating the efficiency of our malicious vehicle traceability scheme.

In conclusion, the experimental results demonstrate that the proposed scheme exhibits excellent efficiency in tracing malicious vehicles, providing a highly scalable and effective solution.

The experimental results show that our scheme enables efficient platoon identity authentication and malicious vehicle tracing. However, due to the use of lattice-based cryptography, integration with existing VANETs infrastructure may be challenging. To address this, a hybrid cryptographic architecture and hardware acceleration, FPGA-based implementations, can be adopted to improve compatibility and practical deployment.

VII. CONCLUSION

In this paper, we propose LETA, a lattice-based efficient and traceable privacy-preserving batch authentication scheme for vehicle platoon in VANETs. The scheme constructs the dynamic VP integrated identity structure VPD-Tree based on hash tree and vehicle pseudonyms to preserve privacy. The VPD-tree and modular lattice are utilized to construct the aggregate signature for secure and efficient batch authentication of VP. Moreover, ZKP is applied on the VPD-tree to trace the malicious vehicles of VP effectively. Formal security analysis and performance experiments are conducted to demonstrate that LETA performs higher security to resist quantum attacks through lattice-based signatures, and takes lower computation and communication costs for the novel VPD-tree structure and efficient batch authentication scheme.

In future work, we will address VP identity authentication in cross-trust-domain scenarios to enhance the secure sharing of parameters and the efficiency of group authentication. Furthermore, by offloading part of the computation to RSUs or MECs, we aim to enable vehicles with limited computational capabilities to meet the requirements of lattice-based cryptography, while simultaneously improving the scalability of the tracing mechanism to achieve load balancing and low-latency performance.

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